NeuroPulse

A Neuroadaptive Companion for Adults Living with ADHD

CMP6213 Mobile & Wearable Application Development

Aathira T Dev 21161041 | Proof-of-Concept Pitch | Birmingham City University 2026

Android | Wear OS | Health Connect | AI Reasoning

The Problem

Why adult ADHD demands a new approach

> *Adults with ADHD do not lack intelligence or motivation — they lack the neurological infrastructure to manage time, transitions, and attention without external scaffolding. Existing tools require them to remember to use the tools(Barkley, 2012).*

01 Time Blindness

Inability to perceive time passing during tasks. Not laziness — a neurological deficit in time perception causing chronic lateness and missed deadlines.

02 Hyperfocus Lock-in

Deep attentional absorption that blocks awareness of external obligations. Flow state becomes a trap. Meetings, calls, and commitments are silently missed.

03 The Interruption Paradox

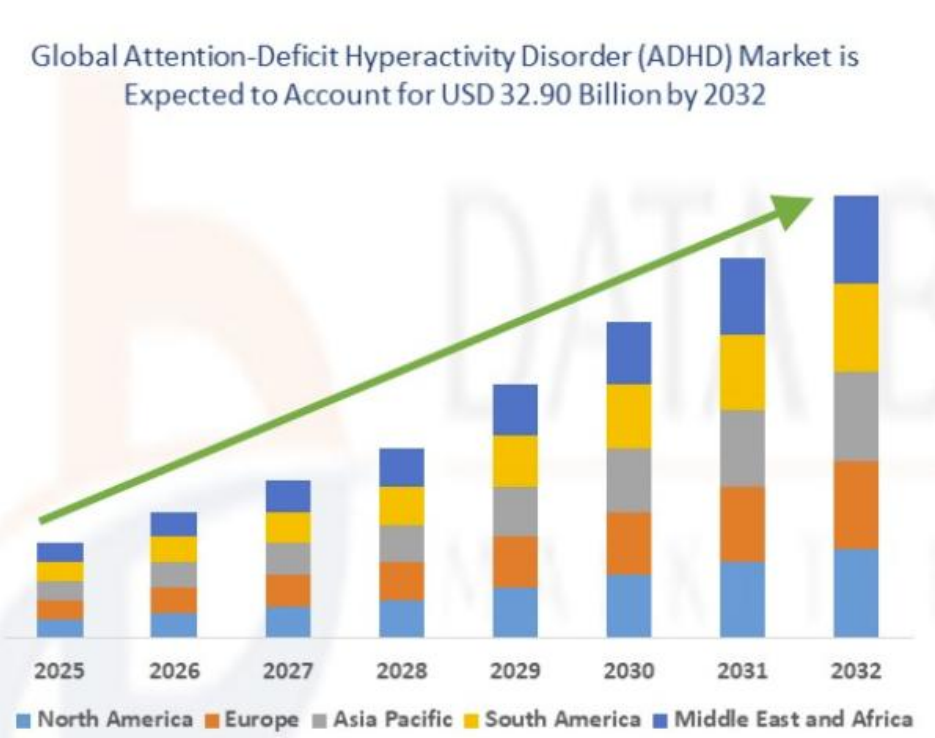
Standard alarms and reminders provide zero lead time — they announce crises, not transitions. The jarring interruption itself causes stress and impaired performance.

04 Tool Friction Problem

Every existing ADHD app requires the user to open it, update it, or act on it first. This assumes the executive function the user lacks is already in place.

Market Research

UK market placement & opportunity(Nice, 2025)



Why Now?

- Wearable adoption surpassed 25% of UK adults (2024)
- Wear OS Health Connect API now enables passive biometric sensing
- NHS Digital lists ADHD as fastest-growing adult referral category
- Adults with ADHD are 3× more likely to use wearable devices
- Employer neuroinclusivity programmes creating enterprise demand
- No market solution combines passive sensing with cognitive scaffolding

User Research

Generative Research — understanding what existing apps lack and what users actually need.

In-depth Interviews

3
participants

NeuroDiverse BCU Society
members recruited via the Discord BCU society network.

What it explored:

- Daily coping strategies and breakdown triggers
- Frustrations with existing ADHD apps (rigidity, no context-awareness)
- Attitude toward physiological monitoring and privacy

Questionnaire

6
responses

Structured self-report survey
to validate themes and gather broader signal beyond the interviews.

What it explored:

- Frequency of context-switching and task abandonment
- Openness to wearable-based mood and stress sensing
- Preferred intervention styles (ambient vs explicit prompts)

Gap Analysis

Identified product requirement gap(Gabarron, Denecke and Lopez-Campos, 2025)

Capability	Focusmate	Tiimo	Goblin Tools	NeuroPulse
Passive background monitoring	✗	✗	✗	✓
Wearable physiological sensing*	✗	✗	✗	✓
ADHD-specific cognitive state detection	✗	Partial	✗	✓
Haptic micro-interventions (watch)	✗	✗	✗	✓
Adaptive task reordering (sleep/HRV)	✗	✗	✗	✓
Zero-friction (no manual input)	✗	✗	✗	✓

*Data Flow : Wearable → Companion App (e.g. Samsung Health, Fitbit) → Health Connect | Typical Sync Behaviour : Samsung Health ~5–15 min during activity, hourly at rest ; Fitbit - On demand or when phone is nearby.

Existing Product Research

Competitive landscape & NeuroPulse novelty

Focusmate *Accountability*	Human-paired focus sessions. No ADHD sensing or wearable support. Requires user to schedule and initiate manually.
Tiimo *Visual Planner*	Structured visual scheduling app. No physiological adaptation or passive sensing. Manual plan updates required.
Goblin Tools *AI Task Helper*	AI-powered task breakdown. Requires deliberate manual input — assumes the initiation capacity ADHD users lack.
Apple Watch Mindfulness *Wellness*	Generic stress reminders. Not ADHD-specific; no contextual intelligence or executive function scaffolding.

NeuroPulse Novelty

- Passive — no manual trigger required
- Wearable-native haptic delivery
- Physiological cognitive state detection
- ADHD-specific intervention logic
- Adaptive to daily sleep & HRV data

The Core Proposition

What NeuroPulse does differently

> *" A wearable-supported cognitive scaffolding system for ADHD executive function failures — that acts before the user knows they need it "*

Passive by Design

No app-opening required. System monitors context continuously and acts autonomously.

Wearable-First Delivery

The smartwatch is the primary output channel — micro-interventions arrive on the wrist.

Physiological Awareness

Heart rate, motion, and sleep data feed a reasoning layer that detects cognitive state.

Two PoC Focus Areas

Time Blindness (Marcus) and Hyperfocus (Zoe) — the two highest-impact ADHD challenges.

User Personas & JTBD

Jobs To Be Done framework(Wilens et al., 2009)

Marcus — Time Blindness

Profile:

- Mid-30s software developer, diagnosed ADHD-inattentive
- Chronically late due to task absorption, not negligence
- Current alarms cause sudden panic — not helpful transitions

Job To Be Done:

"When focused and unable to feel time passing, help me recognise upcoming commitments early enough to transition calmly and arrive on time."

Zoe — Hyperfocus

Profile:

- Late 20s UX designer, ADHD-combined type
- Enters deep focus states unaware of meeting obligations
- Values flow state — abrupt interruptions reduce output quality

Job To Be Done:

"When deeply focused on work, help me surface important commitments without completely breaking my concentration."

MVP Use Cases

Five core interactions demonstrated

01	**Morning Adaptive Day Plan** Trigger: Sleep quality + HRV	Tasks reordered by cognitive capacity	Both
02	**Pre-Event Wrist Briefing** Trigger: Calendar event approaching	Watch shows contextual reminder + time remaining	Marcus
03	**Hyperfocus Alert** Trigger: Extended stillness + wearable motion data	Haptic alert with commitment nudge on watch	Zoe
04	**Stress Grounding Prompt** Trigger: Heart-rate spike detected	Short breathing exercise on smartwatch	Both
05	**Task Completion Reinforcement** Trigger: Task marked complete	Positive micro-affirmation delivered to wrist	Both

UML Use Case Diagram

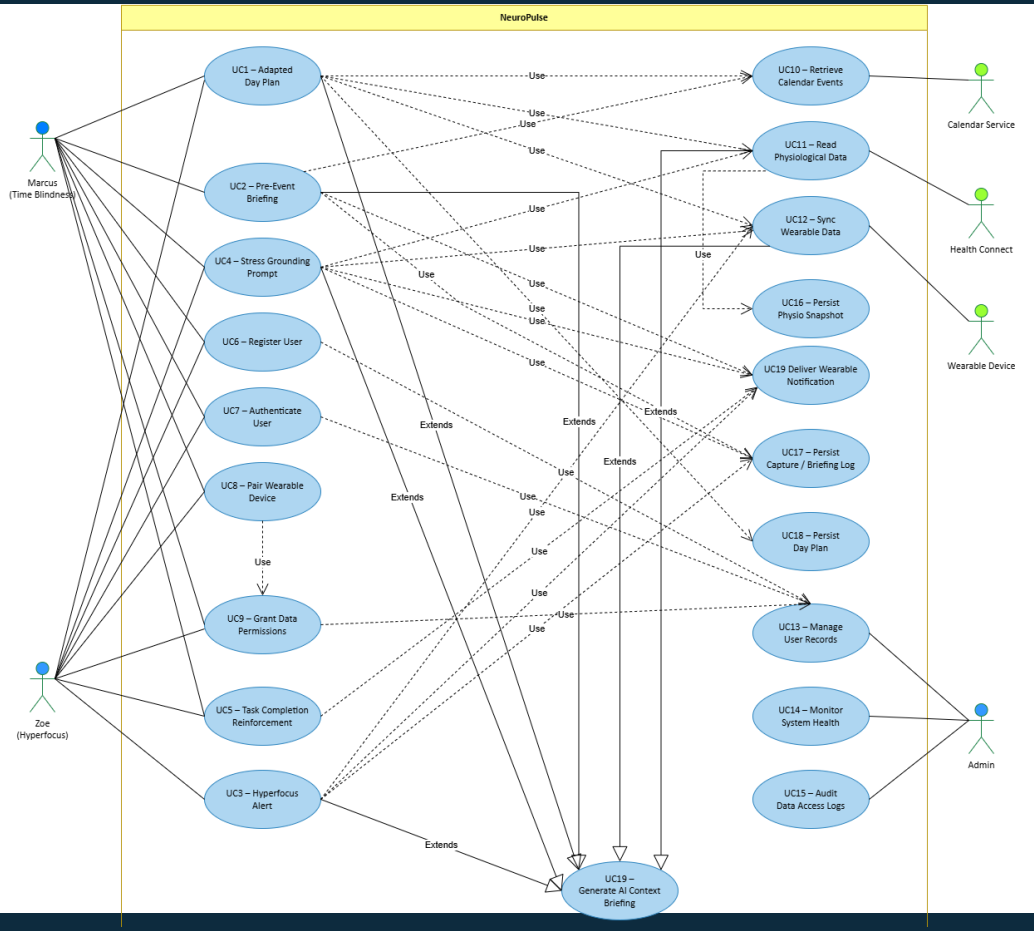
- **Actors reflect real interaction roles** Marcus and Zoe are primary human actors. Calendar Service, Health Connect, and Wearable Device are external systems modelled as *supporting actors* — not use cases — because NeuroPulse interacts with them, not the other way around.

- **`<<include>>` = mandatory, never conditional** UC8→UC9, UC1→physiological data. No `<<include>>` is conditional — every included UC must be available regardless of connectivity, reflecting the offline-first constraint.

- **UC19 is only legitimate `<<extend>>`** UC1, UC2, UC3, UC11, and UC12 all execute fully without UC19. AI context reasoning fires only when capture history meets the minimum threshold — preserving correctness on day one with zero data.

- **Physiological signal chain is fully traceable** Wearable Device → UC12 → `<<extend>>` UC19 → UC4 Health Connect → UC11 → `<<extend>>` UC19 → UC1 The wearable never directly triggers AI reasoning — signals pass through data retrieval use cases, maintaining clean actor boundaries.

- **Persistence is system-owned, never actor-initiated** UC16, UC17, and UC18 are included by operational UCs only — no actor directly initiates a persist operation. Health and behavioral data stays device-local, satisfying GDPR(Voigt, 2019) Article 9 special category data requirements.



App Features

PoC feature set — Android + Wear OS

Android Application

- **Task Planning Interface** — daily task creation with priority weighting
- **Adaptive Task Engine** — reorders tasks based on sleep + HRV data from Health Connect
- **Health Connect Integration** — reads sleep, heart rate, step count passively
- **Room Database** — on-device local persistence (offline-first architecture)
- **Calendar Sync** — reads upcoming events for pre-briefing trigger logic

***AI Context Engine** — RAG pipeline enriches briefings when capture history threshold is met(not user facing)*

Wear OS Companion

- **Contextual Watch Notifications** — pre-event and hyperfocus alerts
- **Haptic Micro-Interventions** — gentle vibration sequences for state transitions
- **Breathing Exercise Tile** — triggered by HR spike detection
- **Task Completion Tile** — positive reinforcement message delivery
- **Sensor Bridge** — forwards motion + HR data to Android logic layer via Wearable Data Layer API

AI Reasoning Implementation ★

Inference over local data — no model training required

Core Principle: *"The model weights are Google's. The context is the user's. RAG injects personal data at inference time — no training, no fine-tuning, no GPU(Asmaidin and Budy Santoso, 2026; Bala, Sharma and Mitchell, 2025)."*

01 OFFLINE MVP Data Collection

PassiveMonitoringClient · ContentResolver · CaptureDao

HR, HRV, sleep scores → physio_snapshots. Calendar events → events table. User notes/photos → captures. All Room — fully offline. Standard Android development — buildable in one week.

02 OFFLINE OUT OF SCOPE Embedding Generation

AllMiniLM-L6-v2 · EmbeddingStore · LangChain4j

Out of scope for PoC — 80MB model load adds startup latency; in-memory store lost on process death. When in scope: AllMiniLM embeddings swapped in behind same CaptureRepository interface, zero upstream changes.

03 OFFLINE MVP RAG Retrieval

CaptureRepository.findRelevant() · Room FTS5 (PoC) → Cosine Similarity (out of scope)

PoC: Room FTS5 full-text search — BM25 keyword ranking, buildable in 2 days. Out of scope: cosine similarity over AllMiniLM vectors. Same method signature — implementation swapped behind the repository contract.

04 CLOUD API MVP LLM Inference

ChatLanguageModel · Gemini Flash API (MVP) → Gemini Nano on-device (out of scope)

Anonymised prompt only — raw sensitive data never leaves device. Free at PoC volumes. Out of scope: Gemini Nano via Android AICore closes connectivity gap. One dependency injection swap — no architectural change required.

AI CONCEPTS IN SCOPE — MVP

RAG — Retrieval Augmented Generation

Personal context retrieved from Room captures at inference time. No population-level training data needed.

FTS5 Full-Text Search

Room's BM25-ranked keyword retrieval engine. Powers CaptureRepository.findRelevant() in PoC. Replaced by cosine similarity when embeddings are in scope.

LLM Inference — not training

Single forward pass through frozen Gemini Flash weights. No backpropagation. No GPU. No training loop required.

Prompt Engineering

BriefingStrategy structures physio context + FTS5 captures into a precise, anonymised instruction for the LLM.

GDPR(Voigt, 2019) — Art.9 Special Category Data

Raw HR, HRV and sleep data never leaves the device. Prompt contains anonymised summaries only. No cloud copy of sensitive data exists in the PoC architecture.

System Architecture

Clean Architecture + Three-Agent Pipeline(Martin, 2017;Murugesan and Igi Global, 2010)

PRESENTATION LAYER

Jetpack Compose UI · ViewModels · StateFlow

↓ *depends on*

DOMAIN LAYER

Use Cases · Repository Interfaces · Pure Kotlin

↓ *depends on*

DATA LAYER

Room DAOs · Health Connect Client · Gemini API

One-Way Dependency Rule

- Domain layer = zero Android dependencies
- Fully testable with JUnit — no emulator needed

Three-Agent Pipeline — Pipe & Filter Pattern

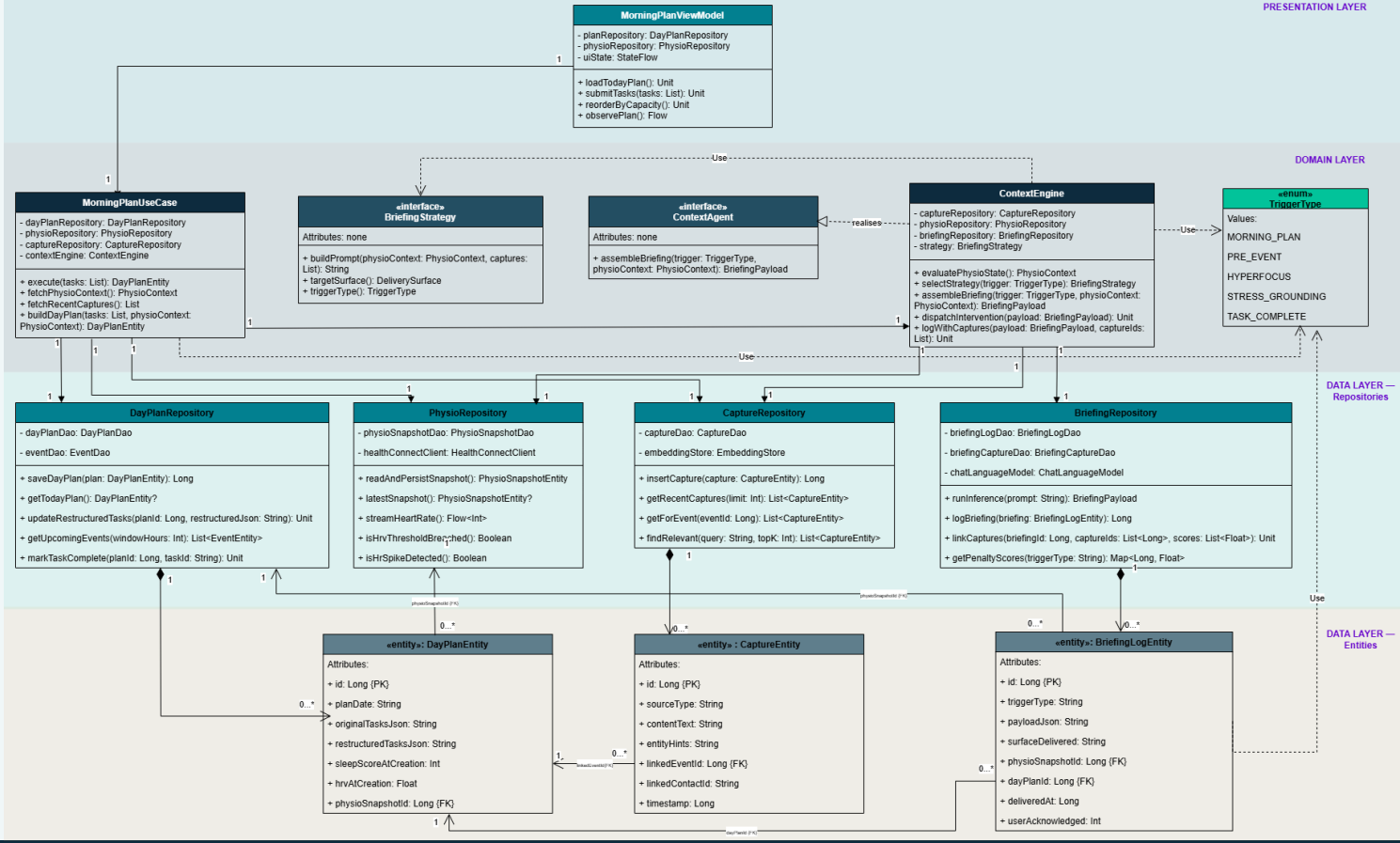
Agent 1 Capture Agent *Data Ingestion Filter*	IN	Sensor events, notifications, photos, calendar
	OUT	Structured rows → Room DB
Agent 2 Context Agent *AI Reasoning Filter*	IN	DB query + PhysioContext + trigger type
	OUT	BriefingPayload (typed Kotlin data class)
Agent 3 Delivery Agent *Routing Filter*	IN	BriefingPayload + surface preference
	OUT	Wear OS tile or Compose notification card

Background Scheduling:
WorkManager (battery-safe, constraint-aware) schedules all background agents. AlarmManager handles exact calendar proximity triggers — WorkManager's 15-min minimum interval is insufficient for 'meeting in 12 min' scenarios.

Class Diagram

Class diagram— key classes, interfaces, and relationships

- **Clean Architecture — One-Way Dependency** ViewModel → UseCase → Repository. Domain layer holds zero Android imports — fully JUnit testable without an emulator.
- **Strategy Pattern for AI Briefings** Each TriggerType gets its own `BriefingStrategy` — prompt structure and delivery surface are decoupled from `ContextEngine`. To accommodate future work.
- **RAG over Fine-Tuning** No model training. Captures are retrieved by semantic similarity at inference time — personal context injected via prompt. Correct architectural choice for a hyper-personal system.
- **TriggerType as First-Class Enum** Five values map 1:1 to UC1–UC5. Every briefing, log entry, and strategy selection is typed — no magic strings anywhere in the pipeline.



Data Persistence — Exploration

Five options evaluated against NeuroPulse's three non-negotiable requirements

Requirements: ① Privacy-first — all data on-device (GDPR(Voigt, 2019)) ② Offline operation — core features work without internet ③ Query flexibility — time, person, event, physiological state

Option	What It Is	Pros	Cons	Verdict
Room (SQLite)	*On-device Relational ORM*	✓ Kotlin Flow. FTS5 full-text search. Compile-time query validation. Migration support. On-device. Offline.	✕ Multi-table join complexity. Schema migrations needed as app evolves.	**CHOSEN**
SQLDelight	*KMM SQL-first library*	✓ Explicit SQL control. Kotlin Multiplatform ready (iOS future expansion).	✕ Less Android-native than Room. Smaller community. Migration tooling less mature.	Rejected
Firebase Firestore	*Cloud-hosted NoSQL*	✓ Real-time sync. Multi-device. Flexible document model.	✕ Cloud-hosted — violates privacy-first requirement. GDPR(Voigt, 2019) complexity. Ongoing cost. Requires internet.	**Considered**
Realm (MongoDB)	*On-device object DB*	✓ Object-oriented. Fast. On-device capable. Optional cloud sync.	✕ Larger footprint. Weaker Kotlin Flow integration. Optional cloud sync re-introduces privacy risk.	Rejected
DataStore (Prefs)	*Jetpack key-value store*	✓ Lightweight. Coroutine-native. Type-safe proto variant. Perfect for settings.	✕ Not a relational DB. Cannot store captures, events, or physio data at required complexity.	**Supplementary (settings only)**

Data Persistence — Proposed Plan

Room as the single source of truth — data type to storage mapping

Principle : “ The intelligence is cloud-assisted but the data is device-sovereign “

Data type	Where	Why
Physiological snapshots	Local (Room)	Trigger fires offline; PhysioSnapshotEntity – UC11
Captures	Local (Room)	RAG retrieval at inference time; CaptureEntity, CaptureRepository.findRelevant()
Calendar events	Local (Room)	Proximity trigger for UC2; DayPlanRepository.getUpcomingEvents()
Day plans & briefing log	Local (Room)	Core operational data
Briefing capture links	Local (Room)	RAG feedback loop — penalizes dismissed briefings; BriefingCaptureEntity
User identity / preferences	Cloud	Login persistence, cross-device
Optional backup	Cloud	User choice, not on critical path
LLM inference (PoC)	Cloud API	Gemini; on-device (Nano) in v2 ; BriefingRepository

Proposed Data Design

Room database schema — Entity-Relationship model

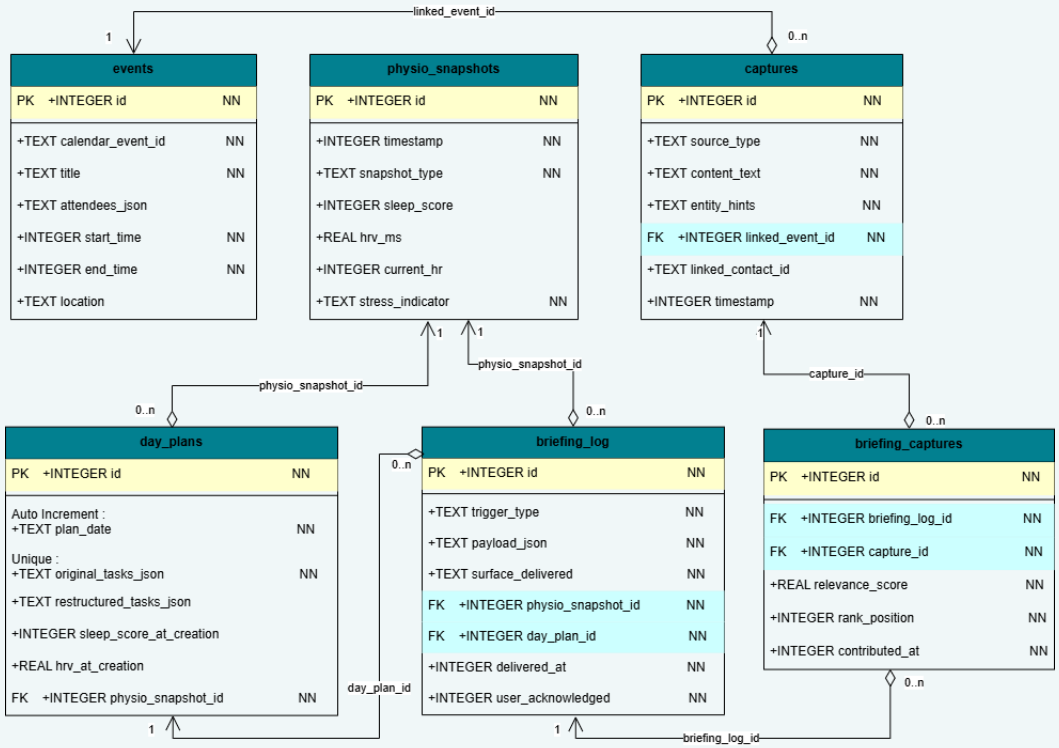
- **Room over Firebase** Offline triggers demand a local source of truth. Firebase's eventual consistency model is architecturally incompatible. GDPR(Voigt, 2019) Art.9 compliance achieved by default.

- **briefing_captures — beyond a join table** Stores `relevance_score` and `rank_position` per briefing. Enables `getPenaltyScores()` — dismissed briefings penalise contributing captures over time.

- **Nullable FKs encode resilience** `physio_snapshot_id` and `day_plan_id` nullable in `briefing_log`. Briefing delivery never blocked by missing context — graceful degradation by design.

- **TriggerType as schema contract** `trigger_type` values map 1:1 to UC1–UC5. Typed enum enforced at application layer — no magic strings in the schema.

- **M:N resolved with attribution** `captures` ↔ `briefing_log` resolved via `briefing_captures`. Every AI briefing is auditable — which captures contributed, in what order, at what relevance score.



Additional Frameworks & APIs

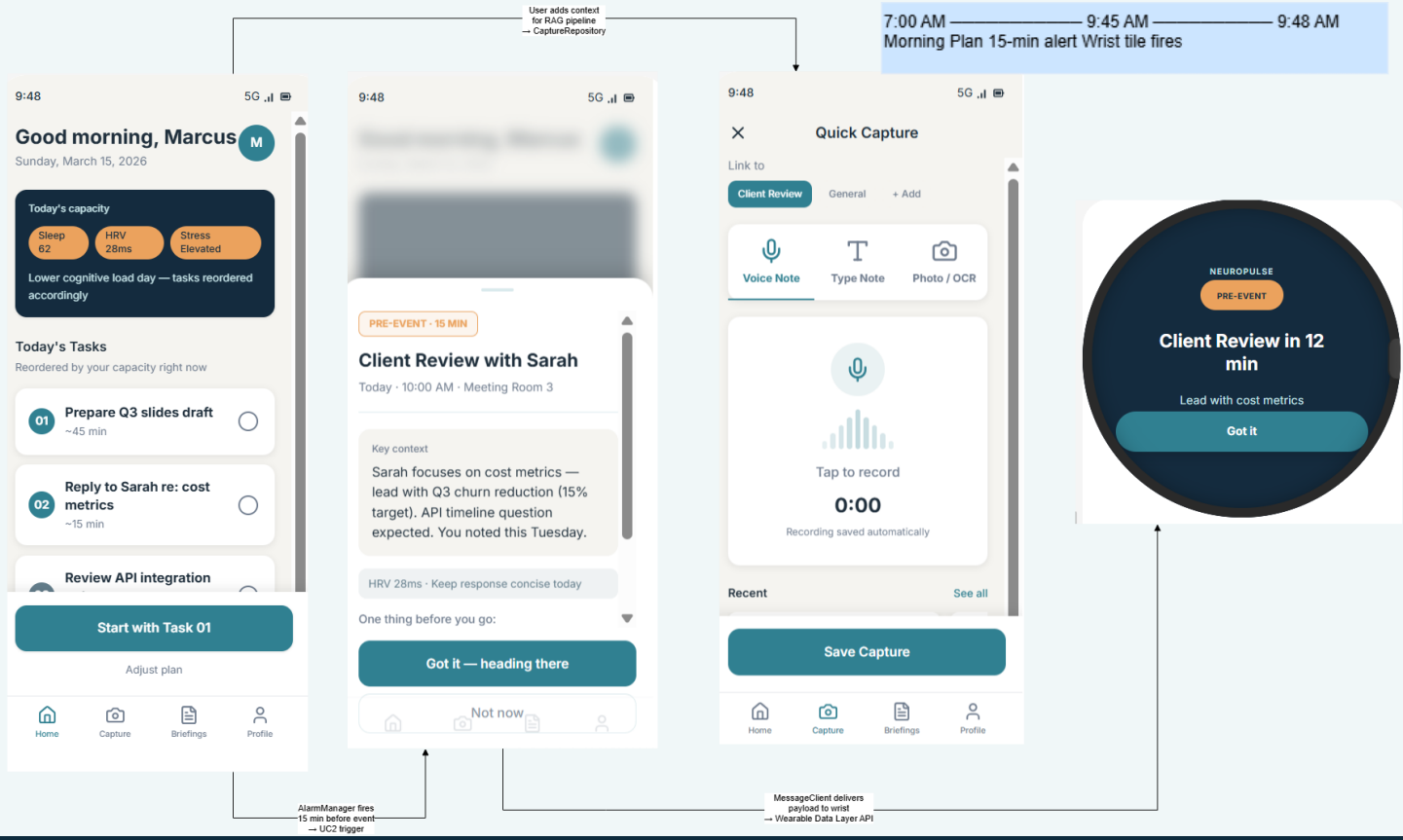
External dependencies — rationale, trade-offs, and alternatives

Framework	Category	Why Selected	✓ Pros	✗ Cons	Alt.
Room (SQLite ORM)	Local Persistence	All 6 ER entities are Room @Entity classes. DAOs return Flow<T> natively. FTS5 full-text search powers CaptureRepository.findRelevant() in MVP. Compile-time @Query validation.	FTS5 built-in. Migration support. Kotlin Flow native. Zero network dependency.	Multi-table join complexity. Schema migrations needed as schema evolves.	*SQLDelight*
Jetpack Compose	UI Framework	Declarative, state-driven UI. Single paradigm across phone & Wear OS. MorningPlanViewModel and BriefingViewModel expose StateFlow — Compose reacts automatically.	Reactive state→UI. Wear tiles native. Less boilerplate.	Steeper learning curve vs XML Views.	*XML Views*
Hilt (Dagger Hilt)	Dependency Injection	Injects DayPlanRepository, PhysioRepository, BriefingRepository into MorningPlanUseCase and ContextEngine. ChatLanguageModel binding swapped for Gemini Nano in v2 with zero code change.	Compile-time safe. Test fakes. Jetpack-integrated.	Annotation processing adds build time.	*Koin*
Kotlin Flow + Coroutines	Async / Reactive	Room DAOs return Flow<List<T>>. PhysioRepository.streamHeartRate() returns Flow<Int>. Entire pipeline reactive — zero polling. ViewModels expose StateFlow to Compose.	Structured concurrency. Cancellation-safe. Room-native.	Backpressure tuning needed for high-freq HR streams.	*RxJava*
Health Connect API	Platform SDK	PhysioRepository.healthConnectClient reads sleep, HRV, HR from Wear OS, Fitbit, Garmin via PassiveMonitoringClient. Maps to UC11. Privacy-controlled by user. Google Fit deprecated.	Cross-device. GDPR(Voigt, 2019)-compliant. Android standard.	Permission required. Unavailable below Android 9.	*Wear OS sensor APIs*
Wearable Data Layer API	Wearable Comms	MessageClient delivers UC2–UC4 briefing payloads to wrist. Watch-side HR spike detection sent back to phone via same channel to trigger UC19. Maps to UC12 Sync Wearable Data.	Bidirectional. BLE-based — no internet needed.	Companion app required on watch. Latency on BLE reconnect.	*Direct Wear OS notification*
LangChain4j	Agent Orchestration	Wraps Gemini Flash behind ChatLanguageModel interface used by BriefingRepository.runInference(). EmbeddingStore interface ready for semantic retrieval when embeddings come in scope.	Model-agnostic. Typed outputs. Apache 2.0.	Adds dependency. Prompt engineering required.	*Direct Gemini API*

Application External Design

UI screenshots and storyboard

One action per screen Every screen has exactly one primary CTA. No competing equal-weight buttons — eliminates decision paralysis.
Physiological context visible Sleep 62 · HRV 28ms · Stress Elevated — shown as amber pills on the morning screen. User sees why their tasks were reordered without reading an explanation.
Wrist as primary delivery Briefing reaches the user at their wrist before they open their phone. Matches research finding — wearables worn daily, receiving no ADHD-specific interventions.
Proactive not reactive No screen requires the user to open the app first. AlarmManager fires UC2. PassiveMonitoringClient fires UC3. The system meets the user at the moment of need.



Perceived Challenges

Technical and product risks — with mitigations

01 Health Connect Permission Model(Android Health Connect, 2025)

HIGH

Problem

Granular per-type permissions. ADHD users more likely to decline complex flows — reducing core sensing capability.

Mitigation

Calm onboarding explaining each permission's value(Denyer et al., 2023). PhysioRepository.isHrvThresholdBreached() returns false safely when HRV null — UC19 fails gracefully, not catastrophically.

02 Wear OS Battery & Sensor Trade-offs

HIGH

Problem

Continuous HR polling + accelerometer conflict with Doze mode and ambient power management.

Mitigation

PassiveMonitoringClient for passive listening — not active polling. setExpedited() on BriefingDispatchWorker bypasses battery constraints for urgent UC2/UC3 triggers.

03 Cognitive State False Positives(Olinic, Stretea and Chereches, 2025)

MED

Problem

Hyperfocus detection (stillness + HR) fires during meetings or legitimate deep work — eroding trust through over-interruption.

Mitigation

Calendar suppression via DayPlanRepository.getUpcomingEvents(). briefing_log.user_acknowledged feeds getPenaltyScores() — dismissed briefings penalise contributing captures over time.

04 Wearable-Phone Sync Reliability

MED

Problem

Wearable Data Layer API is async — Bluetooth disconnection causes intervention timing slip, critical for time-sensitive nudges.

Mitigation

MessageClient for real-time alerts. ChannelClient for larger payloads. Watch-side fallback out of scope for PoC — degraded to WorkManager window delivery.

05 LLM Connectivity Dependency

MED

Problem

BriefingRepository.runInference() calls Gemini Flash API — requires internet. UC19 enrichment fails offline. Conflicts with offline-first architecture claim.

Mitigation

Deterministic fallback builds structured briefing from PhysioContext + FTS5 retrieval without LLM. Out of scope: Gemini Nano via Android AICore closes gap — one ChatLanguageModel binding swap.

06 RAG Cold Start — No Capture History

MED

Problem

CaptureRepository.findRelevant() returns empty on first use. UC19 extend condition never met — new users receive only deterministic briefings.

Mitigation

DatabaseSeeder pre-populates example captures at onboarding. MIN_THRESHOLD set low for PoC demo. Progressive disclosure — system communicates briefings improve with use.

07 SCHEDULE_EXACT_ALARM — Android 12+

MED

Problem

Exact alarm permission required for AlarmManager UC2 triggers. Android 13+ requires manual Settings navigation — high-risk for ADHD users who skip permission flows.

Mitigation

Detect permission absence at onboarding. Fallback to WorkManager 15-min window — UC2 timing degrades but briefing still delivered. In-app prompt with clear value proposition.

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Software Design Patterns

Architectural decisions — patterns and SOLID principles used during design phase

Design Principle: Every architectural decision must reduce cognitive burden on the user. Patterns that add user-visible complexity are rejected; patterns that improve testability and predictability are preferred.

Clean Architecture + MVVM

Entire application

Strict one-way dependency. Domain layer = zero Android deps. Fully unit-testable with JUnit.

SOLID: Single Responsibility

Pipe & Filter

Three-Agent pipeline

Each agent is a filter: defined input → transformation → defined output. Independently testable. New agents addable without touching existing ones.

SOLID: Single Responsibility Open-Closed

Observer (Flow / StateFlow)

All reactive data streams

Room DAOs return Flow<List<T>>. UI always current with no polling. StateFlow in ViewModel ensures Compose collects the latest state automatically.

SOLID: Open-Closed

Strategy

Briefing assembly per trigger type

Each trigger (CALENDAR, HR_SPIKE, HYPERFOCUS, MORNING) has its own BriefingStrategy. New triggers added without modifying ContextEngine core logic.

SOLID: Open-Closed

Repository

All data source access

Insulates domain layer from knowing whether data comes from Room, Health Connect, or remote API. Clean seam for test fakes.

SOLID: Dependency Inversion

Chain of Responsibility

Physiological state evaluation

SleepQualityHandler → HRStateHandler → ActivityStateHandler → StressIndicatorHandler. Each handler enriches PhysioContext independently — new signals added as new handlers.

SOLID: Open-Closed

Facade

Health Connect API wrapping

HealthConnectFacade presents a clean interface — permission checking, error handling, type mapping all hidden. PhysioRepository makes one call.

SOLID: Single Responsibility

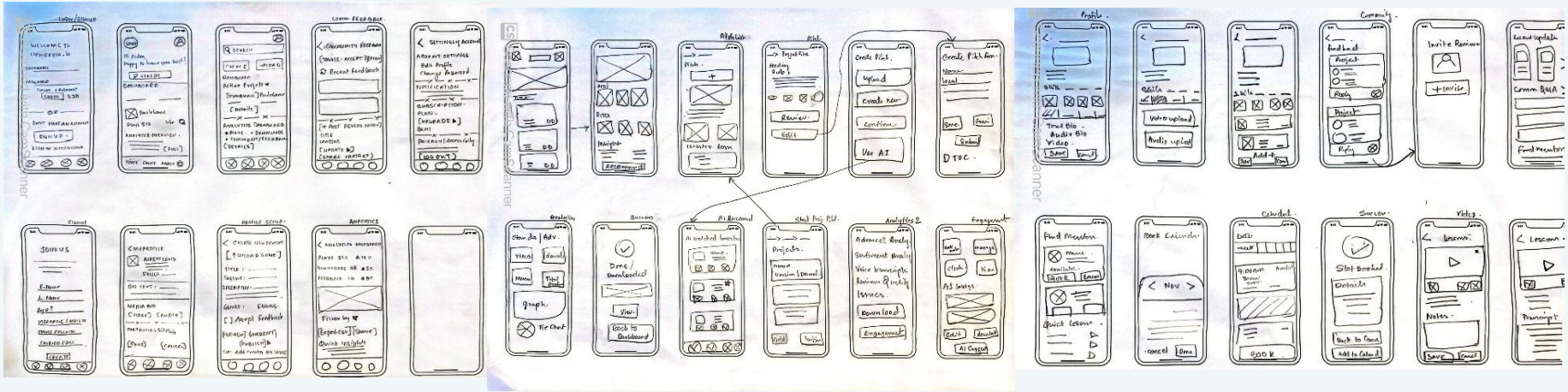
Scheduler (WorkManager)

All background work

Constraint-based scheduling for all Workers. Survives process death and Doze mode. AlarmManager used only for exact calendar proximity (15-min WorkManager gap).

SOLID: Single Responsibility

Low Fidelity Prototype



Expected Impact

User Outcomes

- Reduced anxiety around time transitions
- Fewer missed meetings and commitments
- Improved daily task completion
- Lower cognitive burden — system scaffolds proactively

Clinical Relevance

- Non-pharmaceutical ADHD support layer
- Complements CBT and coaching approaches
- Potential NHS Digital integration pathway
- Session logs enable clinician review

Technical Contribution

- Passive biometric reasoning on Android + Wear OS
- Reference Clean Architecture for neuroadaptive apps
- Advances wearable UX for neurodivergent users
- Scalable to ASD, anxiety, and other executive function domains

NeuroPulse bridges the gap between passive sensing and proactive cognitive support — where no existing product currently does.